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Research Article

Enhanced hepatic metabolic perturbation of polystyrene nanoplastics by UV irradiation-induced hydroxyl radical generation

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ABSTRACT

The environmental behavior of and risks associated with nanoplastics (NPs) have attracted considerable attention. However, compared to pristine NPs, environmental factors such as ultraviolet (UV) irradiation that lead to changes in the toxicity of NPs have rarely been studied. We evaluated the changes in morphology and physicochemical properties of polystyrene (PS) NPs before and after UV irradiation, and compared their hepatotoxicity in mice. The results showed that UV irradiation caused particle size reduction and increased the carbonyl index (CI) and negative charge on the particle surface. UV-aged PS NPs (aPS NPs) could induce the generation of hydroxyl radicals ($\cdot\text{OH}$), but also further promoted the generation of $\cdot\text{OH}$ in the Fenton reaction system. Hepatic pathological damage was more severe in mice exposed to aPS NPs, accompanied by a large number of vacuoles and hepatocyte balloon-like changes and more marked perturbations in blood glucose and serum lipoprotein, alanine aminotransferase and aspartate aminotransferase levels. In addition, exposure to PS NPs and aPS NPs, especially aPS NPs, triggered oxidative stress and significantly damaged the antioxidant capacity of mice liver. Compared with PS NPs, exposure to aPS NPs increased the number of altered metabolites in hepatic and corresponding metabolic pathways, especially glutathione metabolism. Our research suggests that UV irradiation can disrupt the redox balance in organisms by promoting the production of $\cdot\text{OH}$, enhancing PS NPs-induced liver damage and metabolic disorders. This study will help us understand the health risks of NPs and to avoid underestimation of the risks of NPs in nature.

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Introduction

Plastic materials are used practically everywhere and strongly influence our daily lives in many different ways (Lehner et al., 2019; Viitanen et al., 2021; Zhang et al., 2012a). However, the growing production of these materials also leads to an

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increase in plastic waste, which is one of the largest sources of environmental pollution that mankind is exposed to. In the environment, plastic waste can be further broken down into small pieces or particles even smaller than a millimeter in diameter by ultraviolet (UV) radiation, biological degradation, light degradation and embrittlement. Due to the large amount of attention being paid to microplastics (MPs, < 5 mm) released into the environment, several researchers have begun to consider nanoplastics (NPs, < 1 μm), as these plastics, similar to other nanoparticles, may be transported across cell membranes and thus could circulate in the blood and enter organs including the liver, kidney, and even the brain (Zhang et al., 2012b).

Polystyrene (PS) is one of the most frequently used plastics worldwide. It is used in food and industrial packaging, disposable cutlery, medical products and toys (Kumar et al., 2022). Studies have shown that PS NPs can cross the gastrointestinal barrier and deposit in organs such as the gut, liver and kidneys (Deng et al., 2017), causing inflammation and enhancing oxidative stress and lipid metabolism disorders (Estrela et al., 2021; Florance et al., 2021; He et al., 2022; Shi et al., 2023). However, different from pristine NPs, photodegradation of NPs in the environment by the UV portion of sunlight is inevitable, and this radiation is capable of breaking chemical bonds in synthetic polymers. For example, due to the presence of benzene rings, PS can be excited to form a triplet state under UV radiation. The triplet state energy of excited benzene can lead to the decomposition of phenyl groups, which move to the nearest C-H or C-C bond (Yousif and Haddad, 2013), causing cross-linking and oxidation-induced chain scission. This results in decreased particle size, increased specific surface area, and the formation of oxygen-containing functional groups on PS NPs (Liu et al., 2019; Vroom et al., 2017). Generally, the physicochemical characteristics (morphology, structure, polymer type) of NPs correlate with their toxicity (Muller et al., 2008; Zou et al., 2019). During the process of photoaging of a polymer, its internal or external chromophoric groups absorb light and polymerize with oxygen to form polymer alkoxy (PO $\cdot$), polymer peroxy radical (POO $\cdot$) and $\cdot\text{OH}$ (Göpferich, 1996). As one of the most active oxygen free radicals (OFR), $\cdot\text{OH}$ can participate in body metabolism under normal circumstances, while excess $\cdot\text{OH}$ will directly damage cells, cause oxidative stress, and destroy the antioxidant defense function of cells (Srinivas et al., 2019; Villalpando-Rodriguez and Gibson, 2021), thereby enhancing the biological toxicity of NPs. Nevertheless, most of the current studies of aged-NPs have focused on their adsorption of various organic pollutants and heavy metals (Chen et al., 2022; Hüffer et al., 2018), while ignoring the effect of aging itself on their biotoxicity, which may lead to underestimation of their potential health risks.

The liver is a critical hub for numerous physiological processes. These include breaking down xenobiotic compounds, storing glucose, packaging excess lipid, and processing amino acids for energy (Krarup et al., 1975; Trefts et al., 2017). A study of the tissue distribution and accumulation of MPs in mice after oral exposure showed that the liver is the main organ associated with the accumulation of NPs (Deng et al., 2017). In this study, we simulated the aging behavior of PS NPs in nature by UV irradiation, and investigated and compared the hepatotoxic effects of PS NPs and UV-aged PS NPs (aPS NPs)

exposure on mice. In addition, we detected small-molecule liver metabolites by hydrogen nuclear magnetic resonance spectroscopy (^1H NMR), to compare the disturbance to liver metabolic pathways after exposure to PS NPs and aPS NPs. Finally, combined with conventional biochemistry, histological, oxidative damage and antioxidant capacity indicators examination, we analyzed the association between UV-induced $\cdot\text{OH}$ formation and the biological toxicity of aPS NPs. This work reveals the influence of UV irradiation on the biotoxicity of PS NPs, a new perspective on the ecological risks of the aging behavior of NPs caused by environmental factors.

1. Materials and methods

1.1. Preparation and characterization of aPS NPs

The pristine PS NPs (100 nm, Zhongkeleiming, China) suspension was centrifuged at 15,000 r/min for 30 min, the supernatant was discarded, and the PS NPs were alternately washed 3 times with absolute ethanol and double distilled water (DDW) to remove impurities in the stock solution. We obtained aPS NPs by referring to a method reported in the literature with modifications (Cheng et al., 2021). Specifically, the purified PS NPs suspension was diluted with DDW to 5 mg/mL and continuously irradiated with UV light (ZF-1, Kylin-Bell, China) ($\lambda_{\text{max}} = 254 \text{ nm}$) for 14 days, shaking every 12 hr to ensure uniform irradiation. The UV light used in our study could provide 1.64 mW/cm 2 (1.38 mW/cm 2 after absorption by the quartz sample dish) of light intensity as measured by a UV-radiometer (UV-B, Shida photoelectric, China). The surface morphology changes of aPS NPs were observed by a scanning electron microscope (SEM, S-4800, HITACHI, Japan). The surface zeta potentials and hydrated particle size distributions of PS NPs and aPS NPs were characterized using a laser particle sizer (Nano-ZS90, Malvern, China). The chemical bond and functional group changes in aPS NPs were determined by surface attenuated total reflectance infrared spectroscopy (ATR-FTIR, Nicolet iS5, Thermo, USA), and the ratio of the absorbance of the carbonyl peak (1712 cm $^{-1}$) to the methylene peak (1452 cm $^{-1}$) was selected to calculate the carbonyl index (CI) (Rodrigues et al., 2018). Laser Raman Spectroscopy (LRS, XploRA $^{\text{TM}}$ PLUS, HORIBA, Japan) was used for molecular structure analysis; the excitation wavelength was set at 532 nm and the spectrum scanning range was 400–2000 cm $^{-1}$.

1.2. Hydroxyl radical measurement

10 μL of 5-tert-butoxycarbonyl 5-methyl-1-pyrroline N-oxide (BMPO, Sigma, Aldrich) was added to an equal volume of PS NPs and UV-irradiated aPS NPs, respectively. When $\cdot\text{OH}$ was generated, it was rapidly captured by BMPO to form the adduct BMPO-OH, which was subsequently determined using an electron paramagnetic resonance spectrometer (EPR, JES-FA300, JEOL, Japan). To further investigate whether aPS NPs can promote the generation of $\cdot\text{OH}$, a reaction system based on the Fenton reaction ($\text{Fe}^{2+} + \text{H}_2\text{O}_2 \rightarrow \text{OH}^- + \cdot\text{OH} + \text{Fe}^{3+}$) was constructed. 5 μL each of BMPO (259 mmol/L), $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ (1 mmol/L), H_2O_2 (1 wt%) solution and the sample were quickly

mixed, placed in a quartz capillary tube using the siphon principle, and then placed into the resonant cavity of the EPR instrument for measurement. The intensity of the generated BMPO-OH signal was then observed. DDW with or without UV irradiation was used as the control. The working parameters of the equipment were as follows: the magnetic field center was 336 mT, the scanning width was 7.5×1 mT, the modulation width was 0.35×1 mT, the wave amplitude was 3×100 , and the microwave power was 5 mW. The software ultimately determined the relative intensities of individual adducts by automatically fitting experimental spectra.

1.3. Animal experiment

Eighteen male ICR mice (5 weeks old) were provided by SPF Biotech (Beijing, China). Adaptive rearing for one week was conducted with rearing temperature 20–26°C, humidity 40%–70%, and a 12 hr/12 hr light/dark cycle. The mice were randomly divided into 3 groups (PS NPs exposure group, aPS NPs exposure group and control group). After daily weighing, the exposure groups were fed by gavage with PS NPs and aPS NPs suspension at 50 mg/(kg·day), respectively, and the control group was gavage-fed with DDW. All mice were provided with a basic diet throughout the exposure period, with free access to water and food. The water drinking, food intake, appearance and activity were observed throughout the exposure period. On day 28 after exposure, all mice were fasted for 12 hr, blood taken from the eyeballs and then euthanized by dislocation. The mouse liver was quickly removed, and a small piece of the left upper liver lobe was excised, fixed with 4% paraformaldehyde solution and the rest stored in a refrigerator at -80°C. Animal breeding and related experimental procedures met the relevant requirements of the Ethics Committee of Capital Medical University (ethical review number: 2018-0003).

1.4. Blood biochemistry

The 600 μ L of blood was immediately transferred to a sterile centrifuge tube, the sample was centrifuged at 4°C at 1500 r/min for 10 min, and the supernatant was transferred to another centrifuge tube for further analysis. Glucose (GLU), total cholesterol (TC), triglycerides (TG), high density lipoprotein cholesterol (HDL-C) and low-density lipoprotein cholesterol (LDL-C), aspartate aminotransferase (AST) and alanine aminotransferase (ALT) levels in serum were detected using an automatic blood biochemical analyzer (AU680, Beckman, USA).

1.5. Histopathology

The livers of 3 mice in each group were randomly selected for histopathological observation. Briefly, liver tissue fixed in 4% paraformaldehyde solution was dehydrated in ethanol, sectioned, embedded in paraffin, and stained with hematoxylin-eosin (HE) after sectioning. Finally, an automatic section scanning system (Pannoramic scan, 3DHitech, Hungary) was used to scan the liver, and selected images randomly to observe the histopathological image.

1.6. Measurement of antioxidant capacity of liver tissue

The liver can clear exogenous $\cdot$ OH due to its antioxidant system. In this study, we detected the antioxidant capacity of liver by measuring the scavenging ability of liver homogenate for $\cdot$ OH. Specifically, 5 μ L each of BMPO (259 mmol/L), $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ (1 mmol/L), H_2O_2 (1 wt%) solution and DDW were used as blank groups, placed in a quartz capillary tube by the siphon principle, and then placed into the resonant cavity of the EPR instrument for measurement and recording of the $\cdot$ OH signal intensity. Subsequently, DDW was replaced with liver tissue homogenate of each group to observe the ability of liver homogenate to clear $\cdot$ OH. The magnetic field center was 336 mT, scanning width 7.5×1 mT, modulation width 0.35×1 mT, amplitude 1.6×100 , and microwave power 1 mW. The clearance rate of $\cdot$ OH was calculated according to the following formula.

$$\text{OH clearance rate (\%)} = \frac{\text{DDW signal intensity} - \text{experimental group signal intensity}}{\text{DDW signal intensity}} \times 100\%$$

1.7. Measurement of MDA

To 50 mg of liver tissue in each group, normal saline was added in a ratio of 1:9 (mg/ μ L); a 10% tissue homogenate was prepared by a tissue grinder, then centrifuged at 4000 r/min for 15 min at 4°C to detect the content of MDA in the supernatant. The specific operation was conducted according to the instructions of the test kit (A003-1-2, Nanjing Jiancheng Bio-engineering Institute, China).

1.8. Metabolomics sample preparation

To 100 mg mouse liver, 1 mL 80% methanol solution and tungsten carbide beads (4 mm) were added to prepare a homogenate. After centrifugation at 15,000 r/min for 10 min, the supernatant was removed and placed in a rotary evaporator and dried to a yellow gel-like substance. Then, 500 μ L D_2O , 30 μ L PBS (0.1 mmol/L, pH 7.4) and 20 μ L 0.001% trimethylsilyl propionate (TSP) were added and the solution was centrifuged at 15,000 r/min for 10 min at 4°C. 500 μ L of the supernatant was then placed in a 5 mm diameter ^1H NMR tube for subsequent assay.

An ^1H NMR spectrometer (Bruker, Germany) equipped with an ultra-low temperature probe was used for sample detection. The specific parameters were set as follows, the proton resonance frequency was 500 MHz, the pulse sequence noesygpprld (RD-90°- t_1 -90°- t_m -90°) was used for sampling and the temperature was set to 298 K. The number of sampling points in the time domain was 32,000. The spectrum width was 12,000 Hz, the number of accumulations was 128, and the change point was 64,000. The exponential line width factor was 1 Hz, and the phase and baseline were corrected with the TSP peak centered at 0 ppm for chemical shift calibration.

1.9. Data reduction and analysis of ^1H NMR spectra

The human metabolome database (HMDB) combined with some published metabolomic studies was used for metabolite identification in ^1H NMR spectra. Fourier transform, phase

correction, baseline correction and integration of the ^1H NMR spectrum were realized using MestReNova software. The integration area of the spectrum was δ 0.9–9.0, and 0.01 ppm was the integration unit. After integration, the integration value obtained was imported into Microsoft Excel 2019 for quality normalization, and then imported into MetaboAnalyst 5.0 for PCA to obtain the trends in each group. Orthogonal partial least squares discriminant analysis (OPLS-DA) was then performed to compare the PS NPs and aPS NPs exposure groups with the control group, respectively, use R^2Y and Q^2 to examine the validity of the OPLS-DA model. The variable importance projection (VIP) of OPLS-DA analysis was combined with the results of univariate analysis to find the differential metabolites. Finally, we used differential metabolite information for pathway analysis. Impact value > 0.1 and $P < 0.05$ were used as the standards to screen potentially related metabolic pathways.

1.10. Statistical analysis

Statistical analysis was carried out using SPSS 22.0 software. One-way analysis of variance (ANOVA) and the t-test were used to compare differences between groups. The results are expressed as the mean $\pm$ SEM. All of the statistical analyses were two-sided and $P < 0.05$ was considered statistically significant.

2. Results

2.1. Characterization of PS NPs and aPS NPs

SEM images of PS NPs and aPS NPs are shown in Fig. 1a and b. The results showed that the morphology of the PS NPs was uniform and spherical, while the aPS NPs were smaller and

non-uniform, and the morphology was slightly rough. The hydrated particle sizes of PS NPs and aPS NPs were (160.6 ± 0.3) nm and (123.0 ± 1.4) nm, respectively. The zeta potentials were (-30.1 ± 0.1) mV and (-37.2 ± 1.1) mV, respectively. Compared with PS NPs, the hydrated particle size and zeta potential of aPS NPs were reduced, indicating that they were more stable in aqueous solution. The ATR-FTIR spectra of PS NPs and aPS NPs are shown in Fig. 1c, and the absorption peaks of both were similar, in which 3025 cm^{-1} and 1601 cm^{-1} were the stretching vibrations of the C-H and C=C bond on the benzene ring. The peaks at 1452 and 758 cm^{-1} were attributed to the deformation and backbone vibration of C-H. Different from PS NPs, aPS NPs had newly added C=O stretching vibration peaks (1712 cm^{-1}) and O-H vibration peaks of carboxylic acids (3325 cm^{-1}) (Yousif and Haddad, 2013). After 14 days of UV irradiation, the CI increased from 1.06 to 1.90, which indicated that UV irradiation can induce PS NPs to generate oxygen-containing functional groups, which is consistent with previous findings (Zhu et al., 2020). The aPS NPs Raman spectra widened at 1602 cm^{-1} , indicating that the asymmetric stretching vibration between carbon atoms in the benzene ring increased.

2.2. Hydroxyl radical measurement

As shown in Fig. 2, $\cdot\text{OH}$ was captured in the aPS NPs solution formed after UV irradiation, but not in the PS NPs solution or DDW. In the Fenton reaction system, the generation of $\cdot\text{OH}$ significantly increased in the presence of the aPS NPs solution, while the PS NPs solution and DDW before and after irradiation did not achieve the same effect. These results suggested that aPS NPs can not only generate $\cdot\text{OH}$, but also further promote the generation of $\cdot\text{OH}$ in the Fenton reaction system.

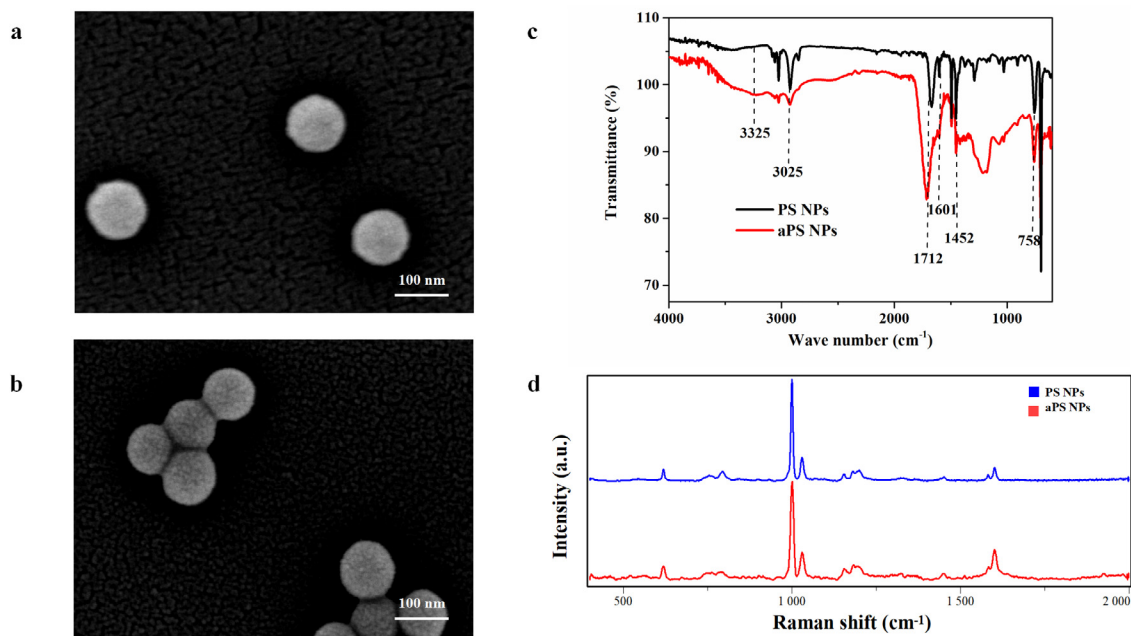


Fig. 1 – Characterization of polystyrene nanoplastics (PS NPs) and aged-polystyrene nanoplastics (aPS NPs). (a) scanning electron microscope (SEM) image of PS NPs; (b) SEM image of aPS NPs; (c) surface attenuated total reflectance infrared spectroscopy spectra of PS NPs and aPS NPs; (d) Raman spectra of PS NPs and aPS NPs.

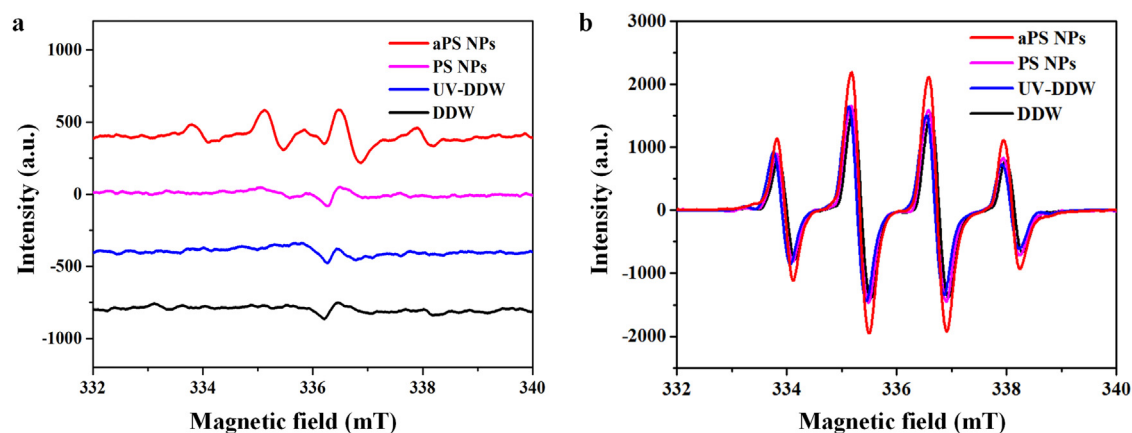


Fig. 2 – Electron paramagnetic resonance (EPR) profile of •OH capture and promotion. (a) EPR profile of •OH capture; (b) EPR profile of •OH promotion.

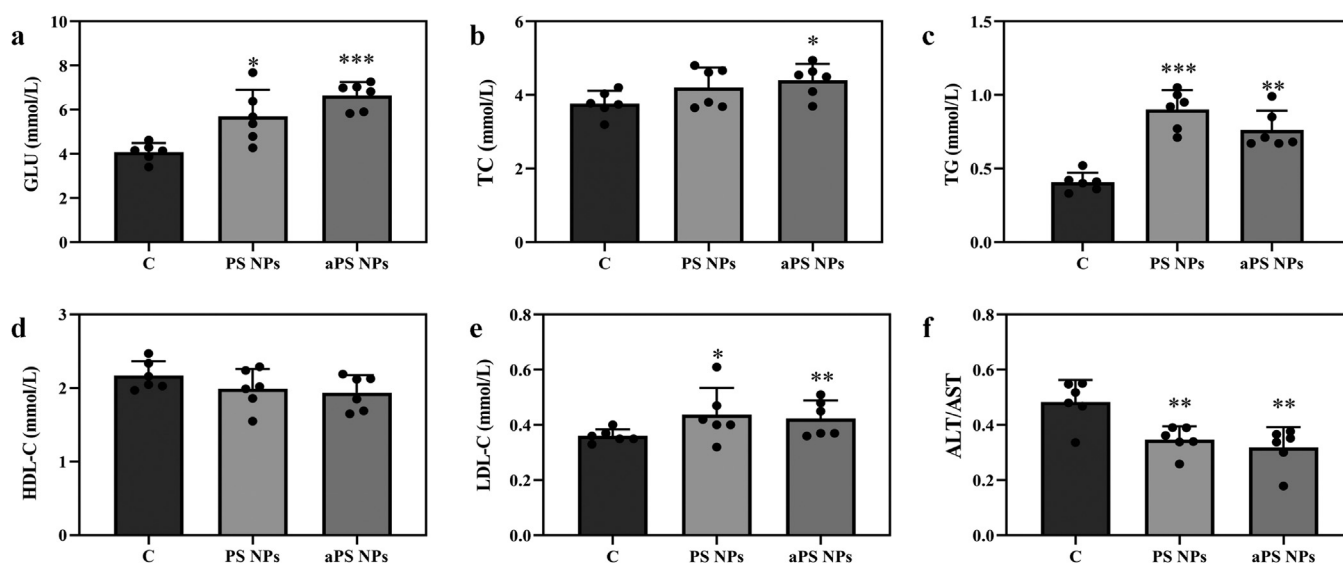


Fig. 3 – Disordered glucose and lipid metabolism in mice following exposure to PS NPs and aPS NPs ($n = 6$). (a) blood glucose (GLU); (b) serum total cholesterol (TC); (c) serum triglycerides (TG); (d) serum high density lipoprotein (HDL-C); (e) serum low density lipoprotein (LDL-C); (f) aspartate aminotransferase (AST)-to-alanine aminotransferase (ALT) ratio. * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$. The scatter points on the bar chart are the original data points.

2.3. Biochemical analysis

No disease symptoms were observed in mice exposed to PS NPs or aPS NPs during the experimental period, and all animals survived to the end of the study. Fig. 3 shows that the blood GLU levels in the PS NPs and aPS NPs exposed groups were significantly higher than those in the control group, and the blood GLU levels in the aPS NPs exposure group were significantly higher than those in the PS NPs exposure group ($P < 0.05$). Compared with the control group, the levels of serum TG and LDL-C in the two exposure groups were significantly increased, and the TC in the PS NPs group showed an increasing trend, while that in the aPS NPs group was significantly increased. Serum HDL-C in both exposure groups showed a

downward trend. ALT/AST were significantly decreased ($P < 0.01$).

2.4. Histopathology analysis

Liver histopathological results showed that the hepatocyte structure of mice in the control group was complete, the nuclei were clearly visible, and the cords were arranged neatly. Hepatocyte pyknosis was observed in the PS NPs- and aPS NPs-exposed groups. The hepatic sinusoids were unclear, cord-like structures disappeared and the intercellular space was blurred, and the central hepatic vein was congested. Liver injury in the aPS NPs group was more severe than that in the PS

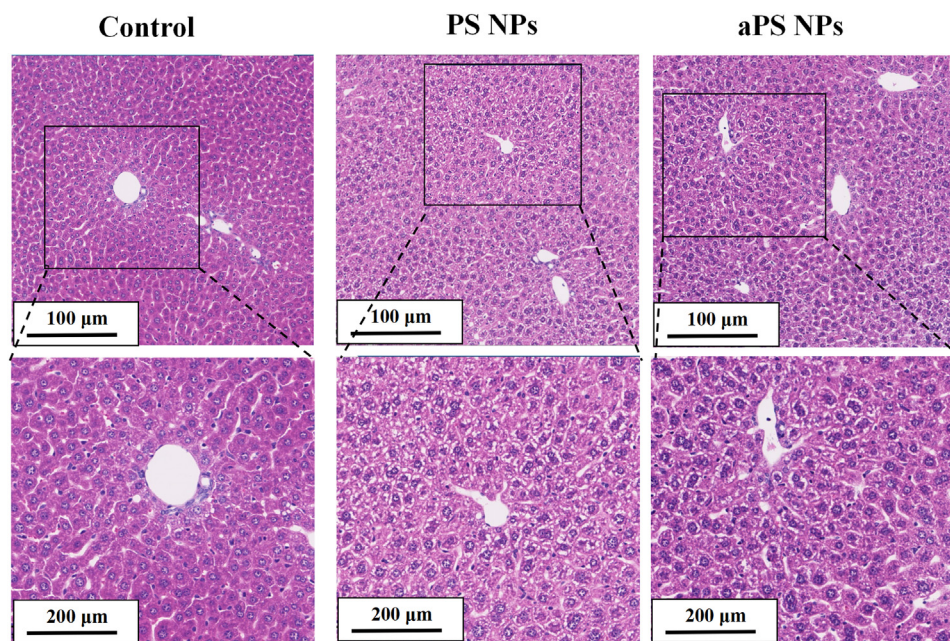


Fig. 4 – The effects of PS NPs and aPS NPs exposure on mouse liver tissue were observed by Hematoxylin-Eosin staining.

NPs group, accompanied by a large number of vacuoles and hepatocyte balloon-like changes (Fig. 4).

2.5. Antioxidant capacity and oxidative damage of liver tissue

EPR results showed that compared with the control group, the scavenging ability for $\cdot\text{OH}$ in liver homogenate of mice in exposed groups was significantly decreased ($P < 0.01$), and the scavenging ability for $\cdot\text{OH}$ in liver homogenate of mice in the aPS NPs-exposed group was significantly lower than that in the PS NPs-exposed group. The results suggested that PS NPs and aPS NPs exposure decreased the antioxidant capacity of mice liver, and the effect of aPS NPs exposure was more significant (Fig. 5a and b).

The results showed that compared with the control group, the contents of MDA in the liver tissue of mice in the PS NPs- and aPS NPs-exposed groups showed an increasing trend, and the increases were more prominent in the aPS NPs-exposed group (Fig. 5c).

2.6. ^1H NMR spectroscopic and pathway analysis

Thirty-four endogenous metabolites were identified in the ^1H NMR spectra of water-soluble liver extract samples (Appendix A Fig. S1), and Appendix A Table S1 lists the chemical shift and content change of each substance. Fig. 6 shows the resulting score plots for PCA (Fig. 6a and b) and OPLS-DA (Fig. 6c and d), respectively. Each point on the score map (points of the same color represent the same group of samples) represents a sample, and the separation between points means that there are individual differences between samples. The results showed that PCA could distinguish the control group from the PS NPs- and aPS NPs-exposed groups. The supervised OPLS-DA was further used to establish a statistical model, and the

analysis results showed that there was a significant separation between the control group and the exposed groups. R^2Y was 0.950 and 0.946, and Q^2 was 0.819 and 0.821, respectively, indicating great model-fitting accuracy (Appendix A Fig. S2). Combining the VIP values and t-test results, cytidine, alanine, acetic acid, fumaric acid, glycine, glutamate, xanthine, creatine, leucine, uridine, lactic acid, trimethylamine, adenosine, valine, acetic acid and isoleucine were significantly decreased compared to the control group. Adenosine triphosphate (ADP) increased in both the PS NPs and aPS NPs groups. In addition, hypoxanthine, choline, succinic acid, and inosine were significantly decreased in the PS NPs group, and cysteine and GLU were significantly decreased in the aPS NPs group.

Metabolites were classified using the online metabolomics analysis website MetaboAnalyst 5.0, and relevant metabolic pathways were screened according to the pathway impact value $F > 0.01$, $P < 0.05$. The results showed that both PS NPs and aPS NPs affected 6 metabolic pathways, including alanine, aspartate and glutamate metabolism, purine metabolism, pyruvate metabolism, glyoxylate and dicarboxylate metabolism, glycine, serine and threonine metabolism, and arginine biosynthesis. Exposure to PS NPs alone affected the tricarboxylic acid (TCA) cycle. aPS NPs also affected glycolysis/gluconeogenesis and glutathione metabolism (Fig. 7). The affected metabolic pathways can be interconnected into a pathway network (Appendix A Fig. S3).

3. Discussion

UV irradiation may lead to chemical bond cleavage in NPs, forming nanopores and oxygen-containing groups that have high photochemical activity and likely form interacting free radicals and ROS intermediates (Hu et al., 2015). This will lead to a decrease in the particle size of NPs and an increase in the

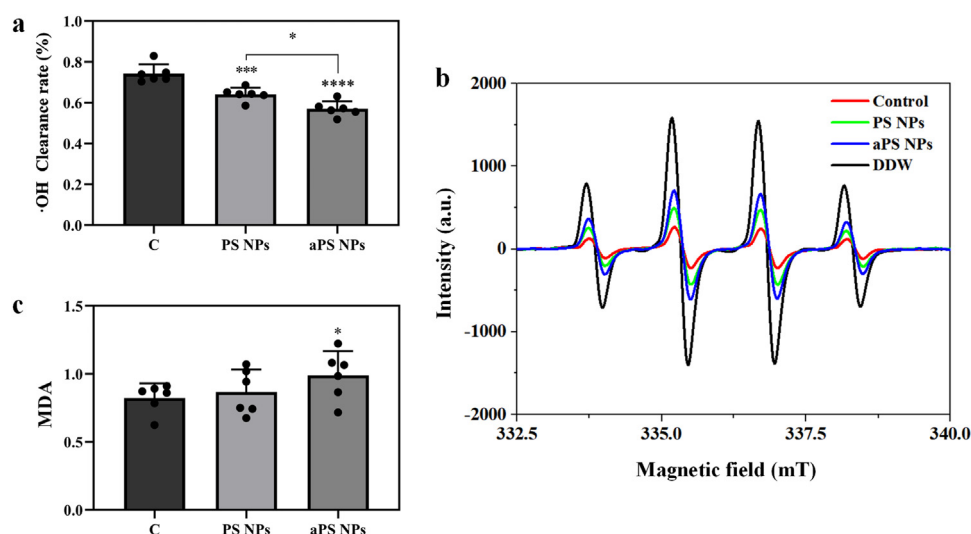


Fig. 5 – Effects of PS NPs and aPS NPs exposure on antioxidant capacity and oxidative stress in mouse liver tissue. (a) $\cdot\text{OH}$ clearance rate; (b) EPR spectrum of $\cdot\text{OH}$ clear (DDW: double distilled water); (c) Malondialdehyde (MDA) level in liver tissue. * $P < 0.05$, *** $P < 0.001$, and **** $P < 0.0001$. The scatter points on the bar chart are the original data points.

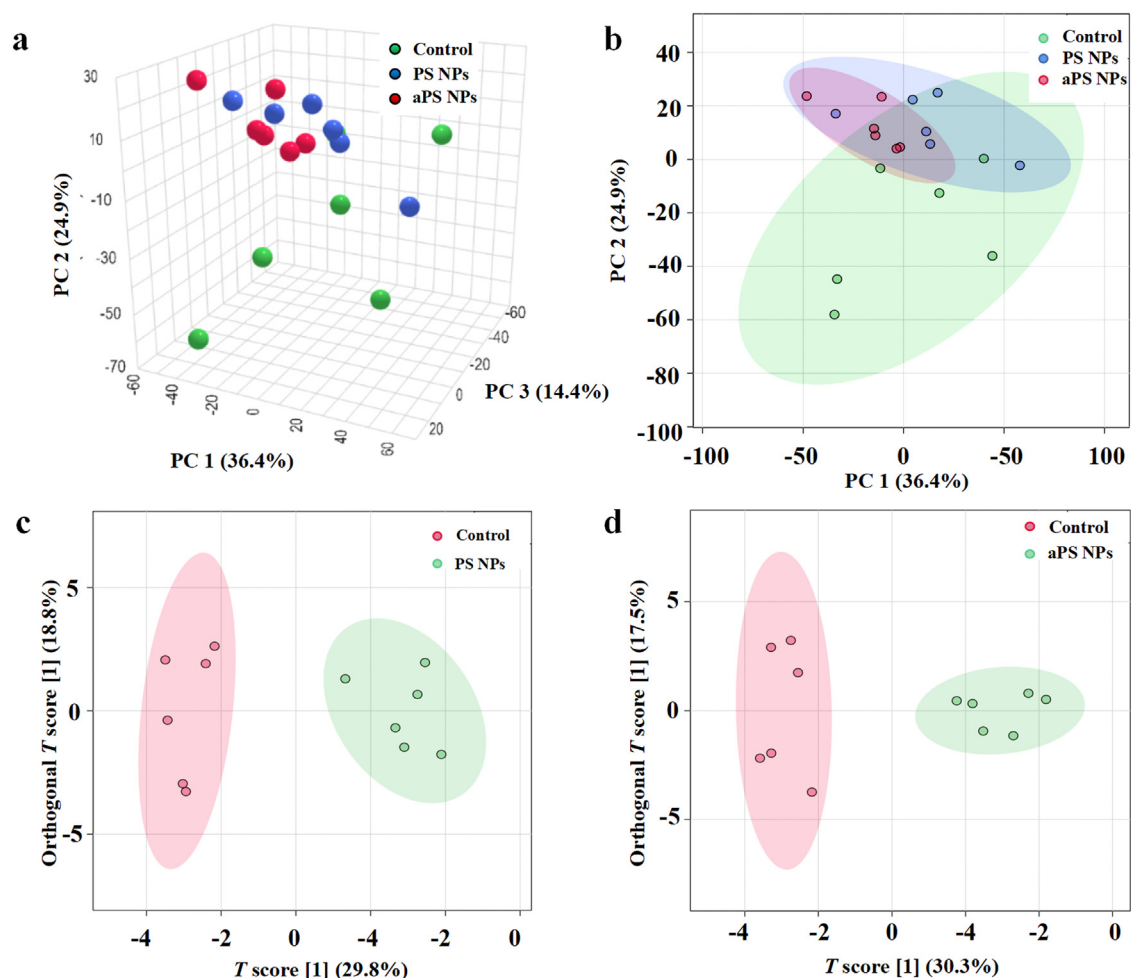


Fig. 6 – Principal component analysis (PCA) and Orthogonal partial least squares discriminant analysis (OPLS-DA) model constructed using hydrogen nuclear magnetic resonance spectroscopy (^1H NMR) data of liver samples. (a) Synchronized 3D plots of PCA; (b) 2D score plots of PCA; (c) OPLS-DA model from control group and PS NP exposure group; (d) OPLS-DA model from control group and aPS NP exposure group.

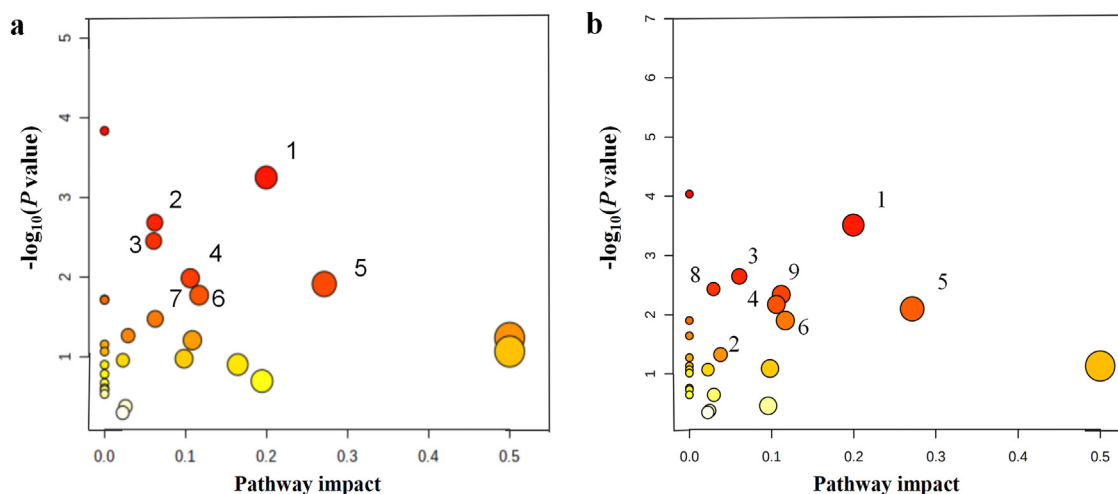


Fig. 7 – Differential metabolite set enrichment analysis. (a) PS NPs vs. control; (b) aPS NPs vs. control. 1. Alanine, aspartate and glutamate metabolism; 2. Purine metabolism; 3. Pyruvate metabolism; 4. Glyoxylate and dicarboxylate metabolism; 5. Glycine, serine and threonine metabolism; 6. Arginine biosynthesis; 7. Tricarboxylic acid cycle; 8. Glycolysis/Gluconeogenesis; 9. Glutathione metabolism.

negative surface charge, which is consistent with our findings. We also observed the formation of a new characteristic C=O vibrational peak in the FTIR spectrum, which was mainly due to C-H bond cleavage caused in PS under UV irradiation, after which carbon combines with oxygen to form a peroxy group (COO·), which adsorbs hydrogen atoms to form a hydroperoxide group (COOH), and is further cleaved to form C=O and ·OH (Yousif and Haddad, 2013). A large number of studies have shown that PS NPs can induce or aggravate damage through ROS-mediated oxidative stress (Tang et al., 2022; Wu et al., 2022a, 2022b). In the process of UV irradiation, ROS formed on aPS NPs mainly include superoxide anion ($\cdot\text{O}_2^-$) and ·OH (Zhu et al., 2019). Under normal physiological conditions, $\cdot\text{O}_2^-$ plays a role in signal regulation as a functional molecule, and toxic effects occur only when its concentration is extremely high or when it is converted to ·OH in the presence of metal ions (Sun et al., 2015). ·OH is a kind of toxic ROS with high activity; the excessive accumulation of ·OH has very prominent toxic effects on the body, inducing lipid peroxidation of cell membranes, damaging proteins, enzymes and DNA (Su et al., 2019; Villalpando-Rodriguez and Gibson, 2021). Therefore, we mainly focused on the effect of aPS NPs on ·OH generation in this study. EPR results confirmed the presence of ·OH in aPS NPs solution, and the radicals and ROS intermediates generated can act as initiators of secondary reactions, further aggravating the generation of ·OH (Yousif and Haddad, 2013). Then we simulated the generation of ·OH in vivo by the Fenton reaction. The results showed that aPS NPs could also promote the production of ·OH, which may explain the more severe pathological damage, antioxidant dysfunction and elevated MDA of the group exposed to aPS NPs.

When liver cells are damaged, AST leaks from cell mitochondria into the blood, resulting in elevated AST/ALT levels in the blood. In this study, increased AST/ALT levels in PS NPs- and aPS NPs-exposed groups suggested liver injury

(Mu et al., 2022). In general, elevated blood GLU and TG are strongly associated with obesity, metabolic syndrome and diabetes (Yoon et al., 2022). Apolipoprotein A is mainly present in HDL-C, which transports TC and TG in the blood to the liver for storage, and has a protective effect on the brain and cardiovascular system. Apolipoprotein B mainly exists in LDL-C, which transfers TC and TG from the liver to the entire body. The increase in LDL-C will cause an increase in TG and TC in serum, which will significantly increase the incidence of nonalcoholic fatty liver and cardiovascular disease (Domanski et al., 2020; Gao et al., 2018). These results showed that both PS NPs and aPS NPs could lead to abnormal GLU and lipid metabolism in mice, and aPS NPs resulted in greater damage to various indicators.

Metabolomic methods can be used to detect the changes in the metabolism of organisms by qualitatively and quantitatively analyzing small molecules in biological samples (Kim, 2021; Talmor-Barkan et al., 2022). As the main site of lipid and amino acid metabolism, when the liver is stimulated by NPs, this will inevitably lead to abnormal metabolites, thereby affecting its normal physiological function. Metabolomics studies have shown that both PS NPs and aPS NPs exposure affect the biosynthesis of various amino acids in mouse liver. Since the liver is the main site for the synthesis of amino acids, the decreased amino acid levels may reflect damage to liver function. In addition, PS NPs exposure also affected the TCA circulation, resulting in decreased fumaric acid, succinic acid and creatine, and increased ADP. Fumaric acid and succinic acid are important intermediates of TCA; their abnormal levels suggested that PS NPs exposure affected the function of TCA and the creatine phospho-creatine shuttle system and disturbed the energy metabolism (Philip et al., 2020). Disturbances to energy metabolism can lead to secondary disturbances in GLU and purine metabolism. These changes may reflect a nonspecific toxic effect of NPs, which we speculate

is due to increased energy requirements result from the absorption, distribution, and elimination of PS NPs (Lei et al., 2008). In the aPS NPs-exposed group, we also observed dysregulation of GLU and glutathione metabolism. GLU metabolism disorders were manifested by markedly elevated blood GLU in mice, while liver α , β glucose and glycogen decreased. Previous studies have shown that decreased insulin sensitivity results in decreased GLU uptake capacity, enhanced gluconeogenesis, and impaired hepatic glycogen synthesis (Leavens and Birnbaum, 2011; Li et al., 2009). Decreased acetate levels and increased lactate levels indicated that aerobic metabolism of GLU in the liver of mice was inhibited and that the glycolytic pathway was enhanced. Furthermore, we found that GSH, glycine and glutamate were significantly reduced in the aPS NPs-exposed group. GSH is the main free radical scavenger, and low levels of GSH impair the antioxidant capacity of tissues (Liu et al., 2022). In the metabolic pathway analysis, the effect of aPS NPs exposure on the GSH metabolic pathway was significant, and this result was also mutually verified with EPR. Therefore, we speculated that the enhanced hepatotoxicity was closely related to the ability of aPS NPs to promote $\cdot\text{OH}$ production.

4. Conclusions

In this research, we found that UV irradiation can alter the physicochemical properties of PS NPs and induce and promote the formation of $\cdot\text{OH}$. Compared with PS NPs, aPS NPs could be used as a stressor to induce the exacerbation of oxidative stress, thus causing an imbalance in the oxidation-antioxidant system in liver tissue accompanied by disturbances in liver glycolipid metabolism. Therefore, we believe that UV irradiation will aggravate PS NPs-induced liver function injury and metabolism disorders. The results of this study should serve as a warning to people regarding the non-negligible role of environmental factors in the toxicity of NPs, and the associated health risks should also be given high importance.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A Supplementary data

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jes.2023.06.030](https://doi.org/10.1016/j.jes.2023.06.030).

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